

GENAI PROJECT

Smart resume parser

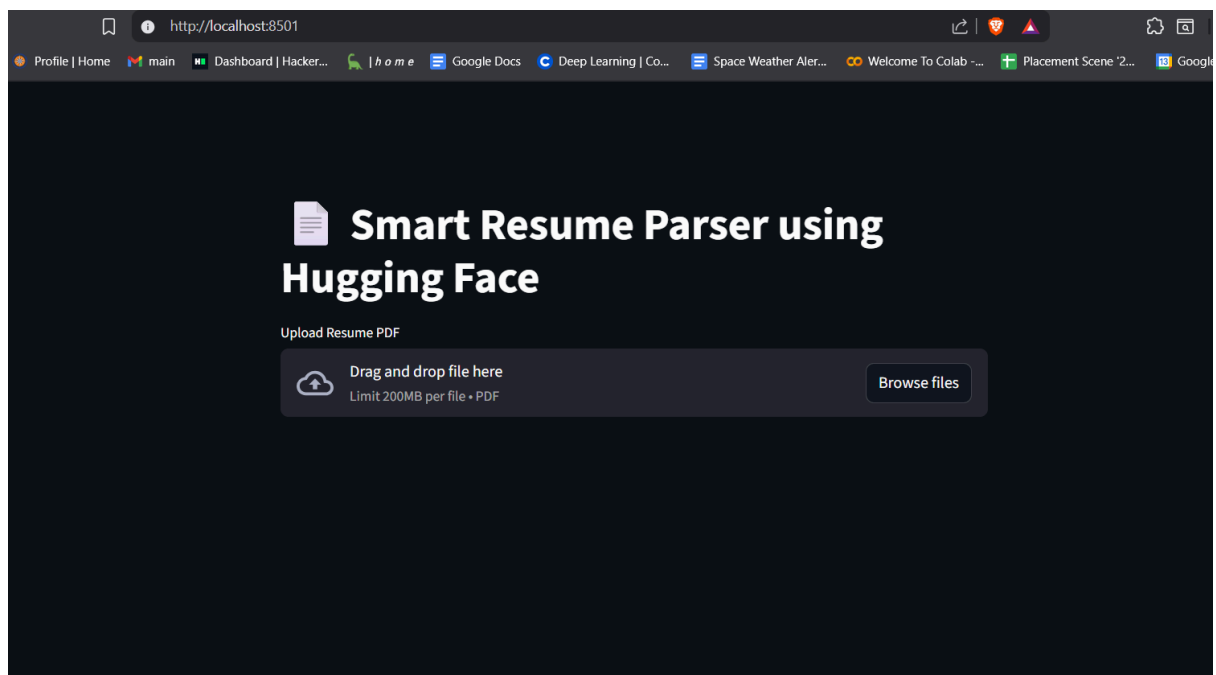
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Abstract

Resumes are unstructured documents that contain valuable information such as personal details, education history, work experience, and skills, but extracting this information manually is time-consuming and error-prone. This project presents a **Smart Resume Parser** that automatically extracts structured information from resumes using transformer-based Natural Language Processing (NLP) models. The system leverages a pretrained Named Entity Recognition (NER) model from the Hugging Face Transformers library to identify and classify entities such as person names and organizations from resume text. The extracted information is processed and converted into a structured JSON format, enabling easier downstream use in recruitment systems and analytics pipelines.

The proposed system also includes a web-based interface built using Streamlit, allowing users to upload resume files in PDF format and view extracted information in real time. The application integrates PDF text extraction, transformer-based entity recognition, and post-processing techniques to enhance the accuracy of extracted fields. This project demonstrates the practical application of foundational NLP concepts such as transformer pipelines, entity extraction, and text preprocessing, and highlights how pretrained language models can be applied to automate real-world information extraction tasks

Output Screenshots





Smart Resume Parser using Hugging Face

Upload Resume PDF



Drag and drop file here

Limit 200MB per file • PDF

Browse files



sample.pdf 125.8KB



Extracted Resume Text

Canton, Georgia 30114
555-555-5555
example@example.com
EDUCATION
Bachelor of Science
|
Mathematics



Parsed Resume Information

EDUCATION

Bachelor of Science

|

Mathematics



Parsed Resume Information

```
{
  "Name" : "Terra #nce Ryan"
  "Organizations" : [
    0 : "G"
    1 : "Group"
    2 : "Georgia College And State University"
    3 : "Cherokee"
    4 : "##yah"
    5 : "##IL"
    6 : "Georgia Professional Standards Commission"
    7 : "##aPSC"
  ]
}
```